

Compiler Design (PCC-CS501)

Course Details

Credit Points: 3

Syllabus Outline

Unit 1

Introduction to Compiling. Compilers, Analysis of the source program, The phases of the compiler, Cousins of the compiler.

Unit 2

Lexical Analysis. The role of the lexical analyzer, Tokens, Patterns, Lexemes, Input buffering, Specifications of a token, Recognition of a tokens, Finite automata, From a regular expression to an NFA, From a regular expression to NFA, From a regular expression to DFA, Design of a lexical analyzer generator (Lex).

Unit 3

Syntax Analysis. The role of a parser, Context free grammars, Writing a grammar, Top down Parsing, Non-recursive Predictive parsing (LL), Bottom up parsing, Handles, Viable prefixes, Operator precedence parsing, LR parsers (SLR, LALR), Parser generators (YACC). Error Recovery strategies for different parsing techniques.

Unit 4

Syntax directed translation. Syntax director definitions, Construction of syntax trees, Bottom-up evaluation of S attributed definitions, L attributed definitions, Bottom-up evaluation of inherited attributes.

Unit 5

Type checking. Type systems, Specification of a simple type checker, Equivalence of type expressions, Type conversions

Unit 6

Run time environments. Source language issues (Activation trees, Control stack, scope of declaration, Binding of names), Storage organization (Subdivision of run-time memory, Activation records), Storage allocation strategies, Parameter passing (call by value, call by reference, copy restore, call by name), Symbol tables, dynamic storage allocation techniques.

Unit 7

Intermediate code generation. Intermediate languages, Graphical representation, Three-address code, Implementation of three address statements (Quadruples, Triples, Indirect triples).

Unit 8

Code optimization. Introduction, Basic blocks & flow graphs, Transformation of basic blocks, Dag representation of basic blocks, The principle sources of optimization, Loops in flow graph, Peephole optimization.

Unit 9

Code generations. Issues in the design of code generator, a simple code generator, Register allocation & assignment.